Google Analytics Integration — Roman Number Converter

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Overview

The Roman Number Converter is a single-page tool that converts integers to Roman numerals and vice versa. Google Analytics 4 was integrated to go beyond passive traffic measurement and capture how users actually interact with the converter, which modes they use, when they succeed, and where they fail.

Live URL: https://juanizn.github.io/RomanNumbers/

Repo: https://github.com/JuanIzn/RomanNumbers/

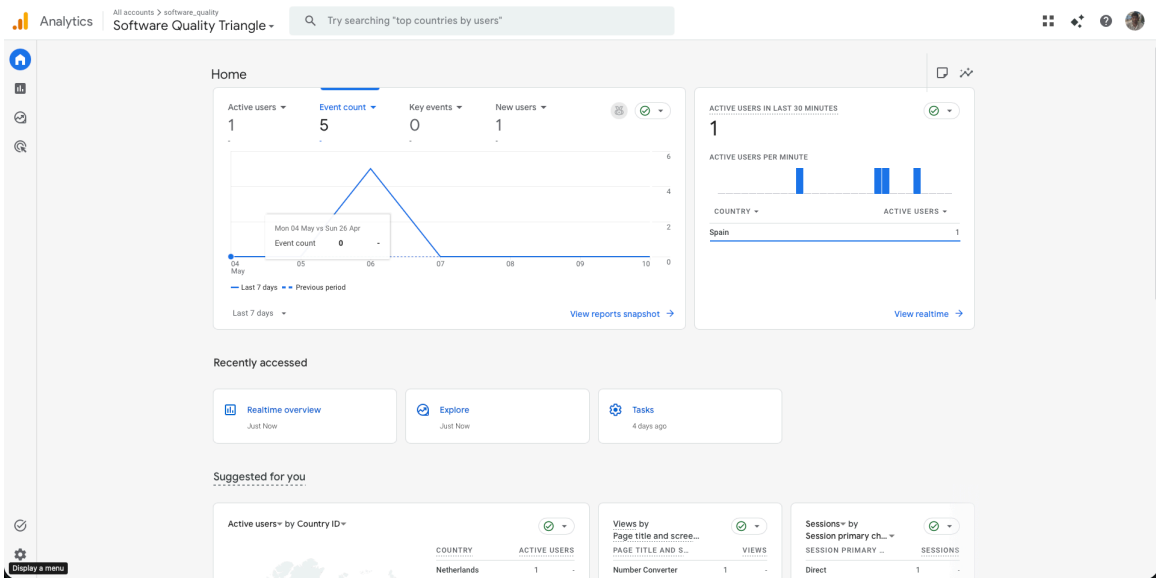

Figure 1: The Roman Number Converter deployed on GitHub Pages

Events Tracked

Three custom events were added beyond the default `page_view`:

Event	Trigger	Parameters
`conversion_success`	User converts successfully	`conversion_mode`
`conversion_error`	Input is invalid or out of range	`conversion_mode`, `error_message`
`mode_changed`	User switches the dropdown	`selected_mode`

`conversion_success` is the core success metric of the tool. The `conversion_mode` parameter reveals whether users prefer Integer → Roman or Roman → Integer — the most actionable insight a converter can produce.

`conversion_error` exposes UX friction. Knowing which error message appears most often (e.g. *"not in canonical form"*) points directly to what needs clearer labeling or a better placeholder.

`mode_changed` distinguishes exploratory users from one-shot users, showing whether the dual-mode design adds value or goes unnoticed.

GA4 Dashboard

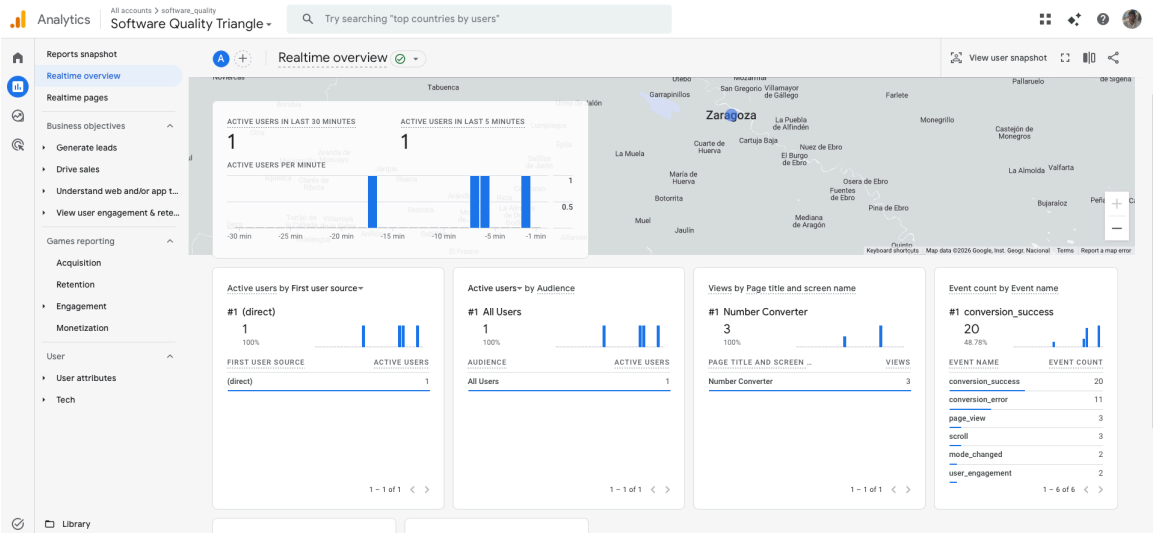

Figure 2: GA4 Events report showing conversion_success, conversion_error and mode_changed

GA4 → Reports → Engagement → Events. All three events appear within minutes of firing. Even at low traffic, DebugView confirms correct event names and parameters on every interaction.